Full Adder with Modes and Dice Roller

Lab 3 and Lab 4 Report

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Authorship Statement

Pranish Parajuli and Gagan Adhikari declare that this lab report is the result of their own original work. Any outside sources used have been properly cited and acknowledged. This report has not been submitted previously for any academic or professional purpose.

We worked together on all sections of the report. The code and flowcharts were created collaboratively. AI tools were used to refine some LaTeX formatting, but all logic and implementation were done manually.

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1 Introduction

This lab was part of the DT124A – Embedded Systems course at HKR. The objective was to design and analyze digital logic systems using fundamental gate-level components and sequential circuits. Lab 3 emphasized the construction of a 3-bit adder and subtractor employing XOR and NAND gates, while Lab 4 investigated data storage mechanisms through SR latches and D flip-flops. A practical application was demonstrated by implementing a dice roller circuit, simulating random number generation between 1 and 6.

2 Materials and Methods

We used the following components and tools:

- Gate ICs (XOR, NAND, AND, OR, NOT)
- D flip-flop IC
- Breadboard with LEDs and jumper wires
- Push buttons
- Function generator for clock signals

Circuits were built on breadboards and tested using manual and generated clock signals. Observations were recorded for each task. For Lab 3, we used fundamental logic gate ICs to build full adders. For bit-wise addition of 3-bit binary numbers, the carry given to the first adder in the series was set to 0, while for subtraction it was set to 1. Each bit of one operand was converted to its 2's complement form using XOR gates to obtain the difference.

For Lab 4, the final task used clock signals, D flip-flops, and logic gates to build a random number generator (from 1 to 6).

3 Results

3.1 Lab 3: Full 3-bit Adder using XOR and NAND Gates

3.1.1 Task 1: Full Adder

A full adder adds three 1-bit inputs: A, B, and Cin. Outputs are Sum and Cout.

Table 1: Truth table for a full adder[1]

A	B	Cin	Cout	Sum
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Figure 1 shows two XOR gates and three NAND gates. Using this circuit combination, a full adder circuit is developed.

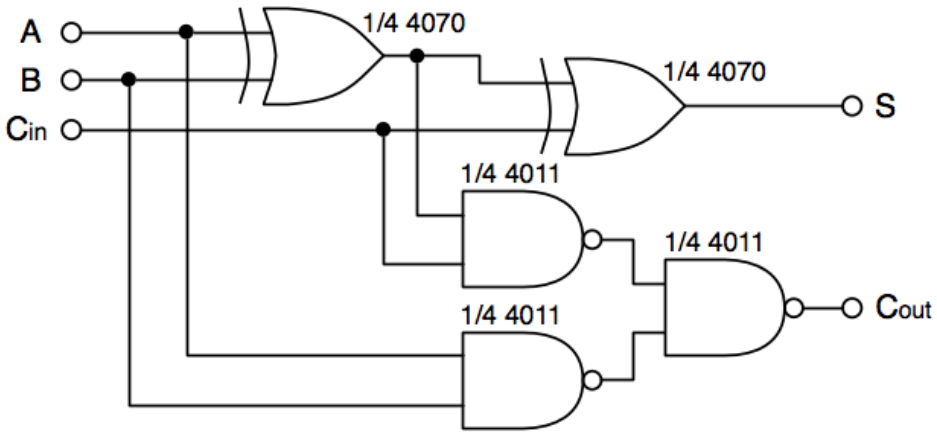


Figure 1: Full adder using XOR and NAND gates. Source: [1]

3.1.2 Task 2: 3-Bit Addition

Figure 2 shows three full adders connected in series. Carry-out from one adder is fed into the next. For a 4-bit output, the carry-out of the last adder is the MSB.

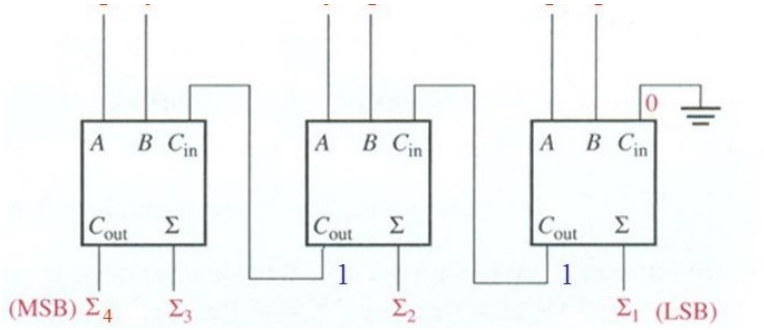


Figure 2: 3-bit adder circuit. Source: [2]

Table 2: Truth table for binary addition using 3 full adders [2]

Operation	Binary Output	Decimal Result
2 + 3	0101	5
5 + 4	1001	9
7 + 1	1000	8
3 + 6	1001	9
5 + 7	1100	12
4 + 6	1010	10

Table 2 shows all possible summation combinations that were tested out after the 3 bit adder circuit was developed in the lab. For example: 2+3 is decimal 5 and its binary in 4 bits is 0101. Here, 0,1,0,1 are $\sum 4$, $\sum 3$, $\sum 2$, AND $\sum 1$ respectively($\sum 4$ BEING THE MSB AND $\sum 1$ BEING THE LSB).

3.1.3 Task 3: Subtraction

Subtraction used 2's complement logic: invert B and add 1. The overflow bit was used to determine if the output is positive or negative.

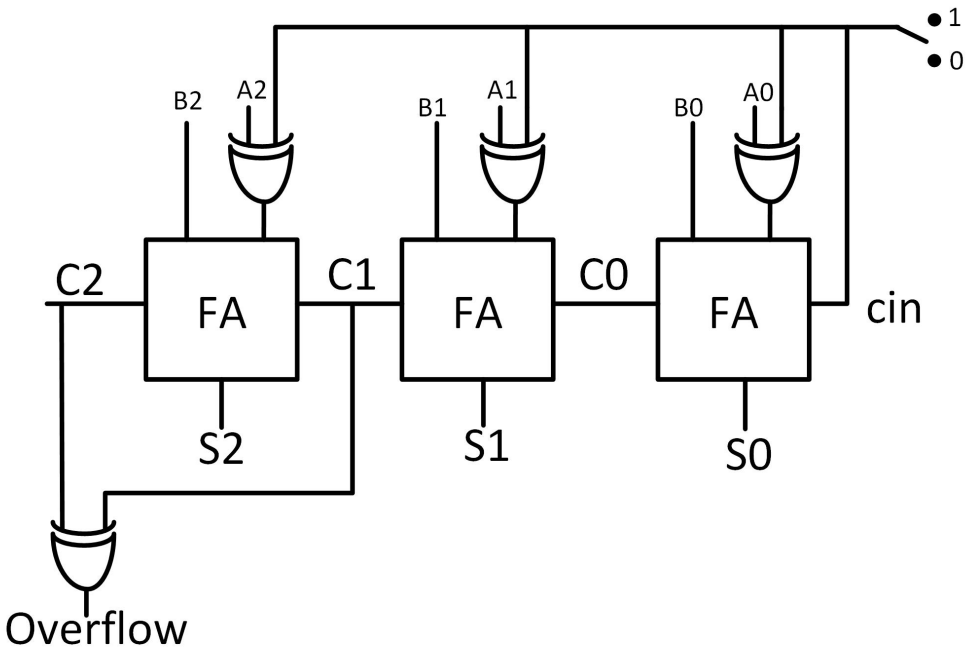


Figure 3: Subtraction using 3-bit adders. Source: [3]

Table 3: Truth table for binary subtraction using three full adders (Basic maths with binary)

Operation	Binary Output	Decimal Result
3 - 2	0001	1
5 - 4	0001	1
5 + 4	1001	9
6 - 3	0011	3
5 - 7	1110	-2
1 - 2	1111	-1

Table 2 shows all possible summation combinations that were tested out after the 3 bit adder circuit was developed in the lab. For example: 3-2 is decimal 1 and its binary in 4 bits is 0001. Here, 0,0,0,1 are Overflow, S2, S1 and S0 respectively(the overflow 0 in here means the output is a positive number).

3.1.4 Discussion

A key challenge encountered during Lab 3 was the propagation of carry signals across chained full adders, which initially led to incorrect outputs. Through systematic debugging and verification of gate wiring, the circuit was successfully stabilized. This exercise reinforced our understanding of arithmetic logic implementation using basic gates and highlighted the role of 2's complement representation in enabling subtraction.

3.2 Lab 4: Latches and Flip-Flops

3.2.1 Task 1: SR Latch

Built using 4011 NAND gate IC. Enabled data storage when EN=0

Table 4: SR Latch Truth Table with Enable Input

Enable (EN)	S	R	Q(next)	Remark
0	X	X	Q(previous)	Latch disabled (no change)
1	0	0	Q(previous)	Hold state
1	0	1	0	Reset
1	1	0	1	Set
1	1	1	Invalid	Forbidden state

From table1 we can see that when EN is 1, the SR latch changes the states, meaning if it was 1 previously it will change to 0 and vice versa. However, if EN is 0, it holds the previous state, and we say it stored the previous value.

3.2.2 Task 2: Dice Roller

Figure 4 shows a circuit made in circuitverse.org by the authors. The circuit uses D flip-flops, AND gates, NOT gates, and OR gates to build a 3-bit counter cycling through 0–6. A button acted as enable; when pressed, the counter advanced with the clock, and when released, it held the dice value.

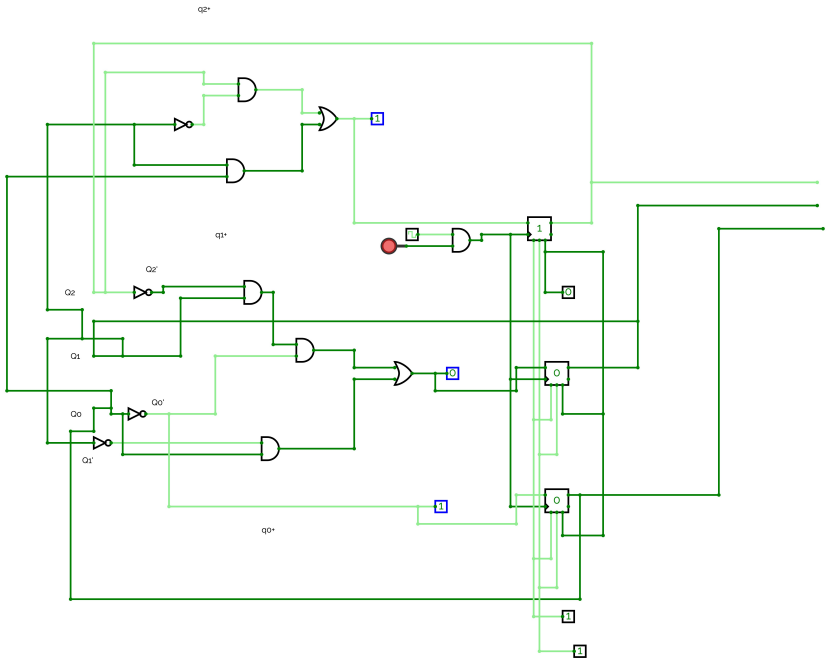


Figure 4: Dice roller using D flip-flops

$$Q2^+ = Q2\overline{Q1} + Q1Q0, \quad Q1^+ = \overline{Q2}Q1\overline{Q0} + \overline{Q1}Q0, \quad Q0^+ = \overline{Q0}$$

3.2.3 Task 3: Random Dice Mode

Clock signal from function generator helped to simulate randomness. We obtained the following results for 20 dice rolls.

Table 5: Observation table for random number generator circuit

Roll	Result	Roll	Result	Roll	Result
1	3	8	1	15	6
2	5	9	2	16	4
3	1	10	3	17	5
4	6	11	5	18	2
5	2	12	4	19	1
6	4	13	2	20	6
7	3	14	1		

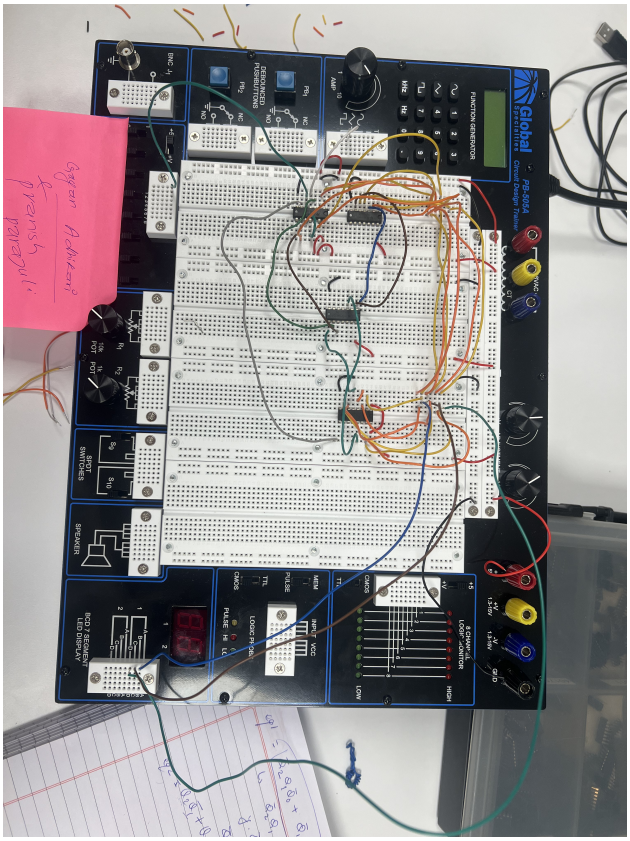


Figure 5: Random number generator using D flip-flops

Figure 5 is a photo taken by the authors. This is the circuit that we built in the lab. Conclusions we got in Table 5 are the results of this particular circuit.

3.2.4 Discussion

We learned how flip-flops store and transition states practically. The dice roller was a good example of a finite state machine. Randomness was limited by clock timing, but results were acceptable. This lab clarified how digital memory works at the most fundamental level.

4 References

References

[1] Github, full adder using XOR and NAND [Online]. Available: https://jacobsa.github.io/circuits/2010/10/02/adders.html

[2] Chegg, 3 bit adder [Online]. Available: https://www.chegg.com/homework-help/questions-and-answers/task-01-3-bit-parallel-adder-found-digital-circuit-figure-1--figure-1-3-bit-parallel

[3] Yilelectronics, Subtraction using 3-bit adder [Online]. Available: https://www.yilelectronics.com/Tutorials/FPGAs/Tutorial_3_Adder_Subtractor/T3

5 Appendix

No code or flowcharts were needed for these tasks.